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Skill Builder : 1

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In [4]: import numpy as np

data = np.load(r"C:\Users\HP\OneDrive\Desktop\CV&DL\student_micro_dataset2.npz")
cifar_x = data['cifar']    # shape: (50, 32, 32, 3)

for i in range(5):
    r, g, b = cifar_x[i][16][16]
    gray = (int(r) + int(g) + int(b)) / 3
    print(f"Image {i+1}: R={r}, G={g}, B={b} → Gray ≈ {gray:.2f}")
```

Image 1: R=245, G=237, B=226 → Gray ≈ 236.00

Image 2: R=218, G=221, B=216 → Gray ≈ 218.33

Image 3: R=112, G=100, B=68 → Gray ≈ 93.33

Image 4: R=181, G=144, B=109 → Gray ≈ 144.67

Image 5: R=92, G=100, B=113 → Gray ≈ 101.67

```
In [5]: import cv2
import numpy as np
import matplotlib.pyplot as plt

data = np.load(r"C:\Users\HP\OneDrive\Desktop\CV&DL\student_micro_dataset2.npz")
mnist_x = data['mnist']    # shape: (50, 28, 28)
cifar_x = data['cifar']    # shape: (50, 32, 32, 3)

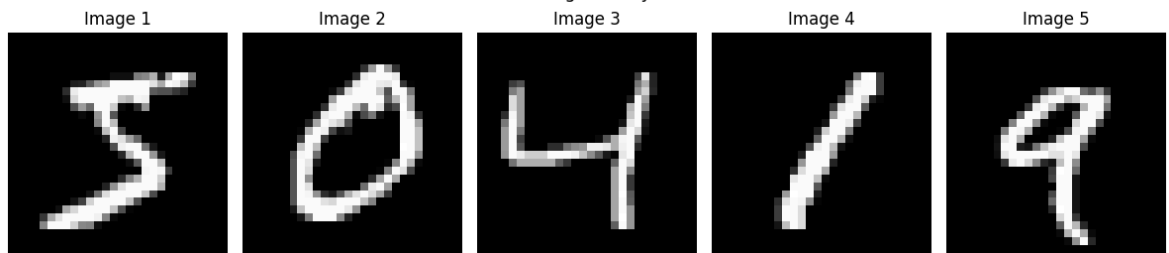
# Display 5 MNIST images
fig, axes = plt.subplots(1, 5, figsize=(12, 3))
fig.suptitle("MNIST Images (Grayscale)", fontsize=13)
for i in range(5):
    axes[i].imshow(mnist_x[i], cmap='gray')
    axes[i].set_title(f"Image {i+1}")
    axes[i].axis('off')
plt.tight_layout()
plt.show()

# Display 5 CIFAR images (RGB)
fig, axes = plt.subplots(1, 5, figsize=(12, 3))
fig.suptitle("CIFAR Images (RGB)", fontsize=13)
for i in range(5):
    axes[i].imshow(cifar_x[i])
    axes[i].set_title(f"Image {i+1}")
    axes[i].axis('off')
plt.tight_layout()
plt.show()

# Convert CIFAR images to grayscale and display
fig, axes = plt.subplots(1, 5, figsize=(12, 3))
fig.suptitle("CIFAR Images (Grayscale)", fontsize=13)
for i in range(5):
    gray = cv2.cvtColor(cifar_x[i], cv2.COLOR_RGB2GRAY)
    axes[i].imshow(gray, cmap='gray')
    axes[i].set_title(f"Image {i+1}")
    axes[i].axis('off')
```

```
plt.tight_layout()
plt.show()
```

MNIST Images (Grayscale)



CIFAR Images (RGB)



CIFAR Images (Grayscale)

